

First, we revised what we studied in the last lecture class, "sinusoidal steady-state response." When a linear circuit is subjected to a sinusoidal input, at steady state, the output of the circuit is sinusoidal with the same frequency. The amplitude of the sinusoidal output is the amplitude of the input multiplied by the magnitude of the sinusoidal transfer function $G(j\omega)$, and the phase of the output is the input phase plus the phase of $G(j\omega)$. Note that the magnitude and phase vary with ω .

We also discussed the DC gain of the transfer function, as well as the last problem in Tutorial 4. Since the circuit is linear, you can apply superposition to find the output.

We also studied how to find state equations when a transfer function is given.

DC Transients

DC transients are temporary shifts in voltage and current that occur in a direct-current circuit before it reaches a stable, steady-state condition. Transients arise because of capacitors (which store energy in an electric field and resist sudden changes in voltage) and inductors (which store energy in a magnetic field and resist sudden changes in current). Purely resistive circuits do not have transients. Transients are triggered by sudden changes such as turning a switch ON or OFF, connecting a power source, or a fault in the network. The duration of a transient depends on the time constant of the system; a circuit generally reaches its new steady state after about 5 time constants.

A General Solution for Step Responses of First-Order Circuits

Consider the following circuits:

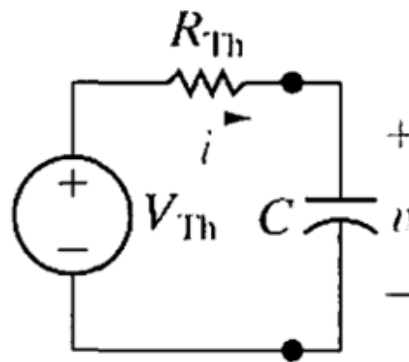


Figure 1: Thevenin equivalent circuit — capacitor voltage $x = v$

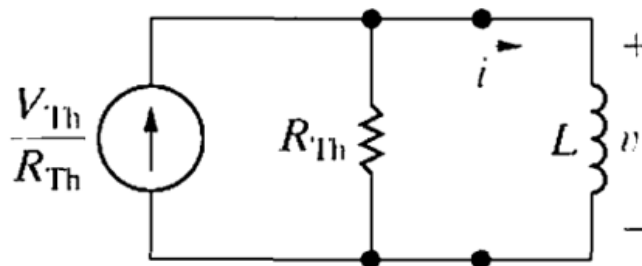


Figure 2: Norton equivalent circuit — inductor current $x = i$

Both circuits can be described by the following general differential equation:

$$dx/dt + x/\tau = K \quad \text{————— (1)}$$

where τ is the time constant of the circuit, and x (which is actually $x(t)$) is the voltage across the capacitor (Figure 1) or the current through the inductor (Figure 2).

If an electrical circuit consists of one energy-storage element (L or C), resistances, and dependent and independent current and voltage sources, it can be replaced by a Thevenin or Norton equivalent circuit, as shown in the figures above.

Solving the Differential Equation

At $t = 0$, $x(0) = x_0$. When $t \rightarrow \infty$, the circuit reaches steady state, so $dx/dt = 0$. Let $x(\infty) = x_f$. Then, from equation (1), $K = x_f / \tau$, and the differential equation becomes:

$$dx/dt + x/\tau = x_f/\tau$$

This equation is solved below using Laplace transforms:

Handwritten solution of the differential equation using Laplace transforms:

$$\begin{aligned} \frac{dx}{dt} + \frac{x}{\tau} &= K \quad x(0) = x_0. \text{ When } t \rightarrow \infty, \text{ the circuit} \\ &\text{will be at steady-state. } \Rightarrow \text{The } \frac{dx}{dt} = 0 \\ \text{Let } x(\infty) &= x_f. \text{ Then } \frac{x_f}{\tau} = K \\ \frac{dx}{dt} + \frac{x}{\tau} &= \frac{x_f}{\tau} \\ \text{Taking Laplace transforms} \\ sX(s) - x_0 + \frac{X(s)}{\tau} &= \frac{x_f}{\tau s} \\ \left(s + \frac{1}{\tau}\right) X(s) &= x_0 + \frac{x_f}{\tau s} \\ X(s) &= \frac{x_0}{s + \frac{1}{\tau}} + \frac{x_f}{\tau s \left(s + \frac{1}{\tau}\right)} \\ &= \frac{x_0}{s + \frac{1}{\tau}} + \frac{x_f}{s} + \frac{-x_f}{s + \frac{1}{\tau}} \\ x(t) &= x_0 e^{-t/\tau} + x_f - x_f e^{-t/\tau} \\ x(t) &= x_f + (x_0 - x_f) e^{-t/\tau} \end{aligned}$$

This gives the general solution:

$$x(t) = x_f + (x_0 - x_f) e^{-t/\tau}$$

Here, x_f is the Thevenin voltage or Norton current, and the time constant τ is $R_{Th}C$ for the capacitor circuit (Figure 1), or L/R_{Th} for the inductor circuit (Figure 2). $x(t)$ is the capacitor voltage $v_C(t)$ or inductor current $i_L(t)$.

Steps:

1. First find x_0 : Draw a circuit before $t < 0$, when the circuit was in steady-state. x_0 is the open circuit voltage across the capacitor (or short circuit current through the inductor).
2. To find x_f and τ : Draw another circuit after $t > 0$, when the circuit is in new steady-state. Just remove the capacitor (or inductor), the circuit excluding the capacitor (or inductor) must be replaced by voltage source V_{TH} in series with R_{TH} (or a current source I_N in parallel with R_{TH}). V_{TH} (or I_N) is x_f and τ is $R_{TH} C$ (or L/R_{TH}). $x(t)$ is the capacitor voltage $v_C(t)$ (or inductor current $i_L(t)$)

Worked Examples

Example 1: RL Circuit

Problem: A series circuit has a 24 V DC source, a 6 Ω resistor, and a 2 H inductor, all in series with a switch. The switch has been open for a long time and is closed at $t = 0$. Find the inductor current $i_L(t)$ for $t \geq 0$.

Before $t = 0$, the switch is open, so no current flows: $i_L(0^-) = 0$ A. Since inductor current cannot change instantly, $i_L(0^+) = 0$ A.

A long time after closing the switch, the inductor behaves like a short circuit, so the current is limited only by the resistor: $i_L(\infty) = V/R = 24/6 = 4$ A.

Only one resistor (6 Ω) is seen by the inductor, so $R_{Th} = 6 \Omega$, and $\tau = L/R_{Th} = 2/6 = 1/3$ s ≈ 0.333 s.

Using $x(t) = x_f + (x_0 - x_f)e^{-t/\tau}$:

$$i_L(t) = 4 + (0 - 4)e^{-t/0.333} = 4(1 - e^{-3t}) \text{ A}, \quad t \geq 0$$

The current starts at 0 A and rises smoothly to 4 A, reaching about 63% of its final value after one time constant ($t = 0.333$ s), and is essentially at 4 A after about $5\tau \approx 1.67$ s.

Example 2: RC Circuit

Problem: A capacitor of 100 μF is initially charged to 50 V. At $t = 0$ it is connected across a 5 k Ω resistor (the source is disconnected, so the capacitor discharges through the resistor only). Find $v_C(t)$ and the current $i(t)$ for $t \geq 0$.

The capacitor is already charged to 50 V just before $t = 0$, and capacitor voltage cannot change instantly, so $v_C(0^+) = 50$ V. There is no source in the circuit, so the capacitor discharges fully: $v_C(\infty) = 0$ V.

The resistor seen by the capacitor is $R_{Th} = 5$ k Ω , so $\tau = R_{Th}C = 5000 \times 100 \times 10^{-6} = 0.5$ s.

$$v_C(t) = 0 + (50 - 0)e^{-t/0.5} = 50e^{-2t} \text{ V}, \quad t \geq 0$$

The discharge current follows from $i(t) = C dv_C/dt$:

$$i(t) = 100 \times 10^{-6} \times (-50 \times 2e^{-2t}) = -5 e^{-2t} \text{ mA}, \quad t \geq 0$$

The negative sign shows that current flows in the direction of discharge, opposite to the assumed charging direction. After one time constant ($t = 0.5 \text{ s}$), the voltage has fallen to about 37% of 50 V, i.e. $\approx 18.4 \text{ V}$; after about $5\tau = 2.5 \text{ s}$ the capacitor is essentially fully discharged.

Example 3: RL Circuit with a Dependent Source

Problem: The switch in the circuit below has been open for a long time before closing at $t = 0$. Find $i_L(t)$ for $t \geq 0$.

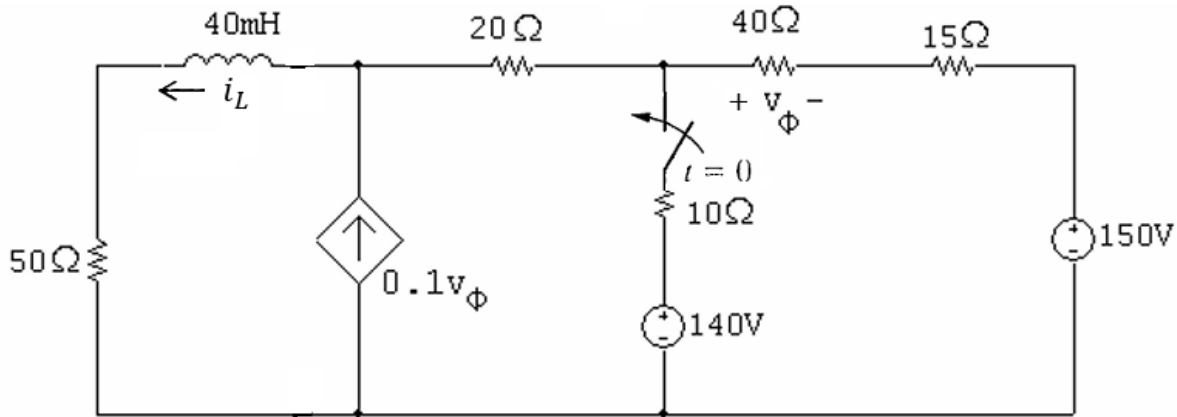
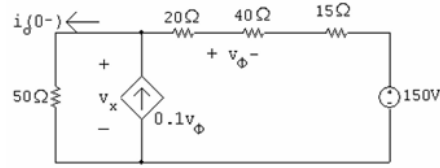


Figure 3: Circuit for Example 3.

Here x is the inductor current. To find x_0 , draw the circuit for $t < 0$. At that time the circuit is in steady state and the inductor is short-circuited; the current through the short is x_0 . For $t > 0$, once the circuit reaches steady state again, draw another circuit and determine I_N and R_{Th} across the inductor terminals. I_N is the short-circuit current through the inductor terminals.

Because the circuit contains a dependent source, R_{Th} cannot be found simply by combining resistors — it must be found by applying a 1 A test source at the terminals, as shown below. The circuit for $t < 0$ (giving $i_0(0) = 6 \text{ A}$) and the circuit and calculation for R_{Th} (giving $R_{Th} = 74 \Omega$) are shown in Figure 4.

For $t < 0$



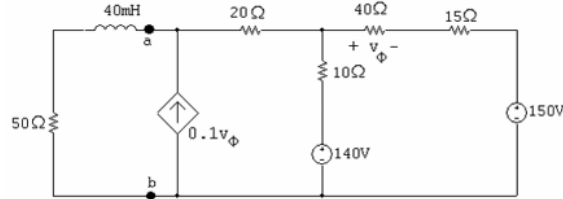
$$\frac{v_x}{50} - 0.1v_\phi + \frac{v_x - 150}{75} = 0$$

$$v_\phi = \frac{40}{75}(v_x - 150)$$

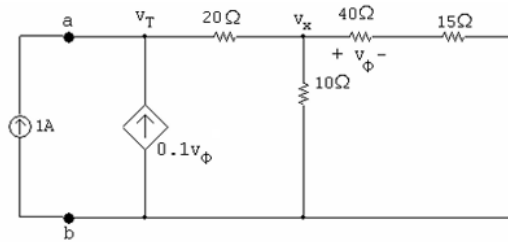
$$\text{Solving, } v_x = 300 \text{ V;}$$

$$i_o(0) = \frac{v_x}{50} = 6 \text{ A}$$

$t > 0$



Find the Thevenin's equivalent resistance with respect to a,b. Apply a 1 A source



$$-1 - 0.1v_\phi + \frac{v_T - v_x}{20} = 0$$

$$\frac{v_x - v_T}{20} + \frac{v_x}{10} + \frac{v_x}{55} = 0$$

$$v_\phi = \frac{40}{55}v_x$$

$$\text{Solving, } v_T = 74 \text{ V} \quad \text{so } R_{Th} = \frac{v_T}{1 \text{ A}} = 74 \Omega$$

Find the open circuit voltage with respect to a, b:

$$-0.1v_\phi + \frac{v_{Th} - v_x}{20} = 0$$

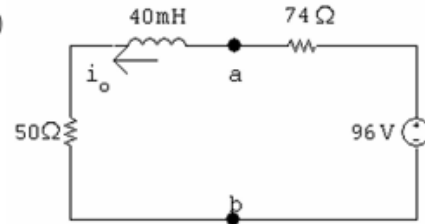
Figure 4: Steady-state circuit for $t < 0$ (top), and the test-source circuit used to find R^{Th} (bottom).

Next, the open-circuit (Thevenin) voltage v_{Th} is found from the circuit with the 150 V source active and the inductor branch open, giving $v_{Th} = 96 \text{ V}$. The final circuit, with the inductor replaced by its 40 mH value and the network reduced to R_{Th} , is shown in Figure 5, along with the final calculation of $i_o(\infty)$, τ , and the complete response.

$$\frac{v_x - v_{Th}}{20} + \frac{v_x - 140}{10} + \frac{v_x - 150}{55} = 0 \quad v_\phi = \frac{40}{55}(v_x - 150)$$

Solving,

$$v_{Th} = 96 \text{ V}$$



$$i_o(\infty) = 96/124 = 0.774 \text{ A} \quad \tau = \frac{40 \times 10^{-3}}{124} = 0.3226 \text{ ms; } 1/\tau = 3100$$

$$i_o = 0.774 + (6 - 0.774)e^{-3100t} = 0.774 + 5.226e^{-3100t} \text{ A, } t \geq 0$$

Figure 5: Calculation of v^{Th} , the reduced circuit, and the final solution for $i_o(t)$.

With $i_o(0) = 6$ A, $i_o(\infty) = 96/124 = 0.774$ A, and $\tau = (40 \times 10^{-3})/124 = 0.3226$ ms, the complete response is:

$$i_o(t) = 0.774 + (6 - 0.774)e^{-3100t} = 0.774 + 5.226e^{-3100t} \text{ A, } t \geq 0$$

Key Points to Remember

A first-order circuit has exactly one capacitor or one inductor, together with resistors and sources. Capacitor voltage and inductor current are the two quantities that cannot change instantly — always start the solution by finding these. Every first-order step response, however complicated the resistive network, follows the same form:

$$\mathbf{x}(t) = \mathbf{x}_f + (\mathbf{x}_0 - \mathbf{x}_f) e^{-t/\tau}$$

For an RC circuit, $\tau = R_{Th}C$; for an RL circuit, $\tau = L/R_{Th}$. When the circuit contains a dependent source, R_{Th} must be found by applying a test source (1 A or 1 V) at the terminals rather than by simple resistor combination. After about 5τ , the transient has essentially died out and the circuit has reached its new steady state.